

ASUS ROG STRIX X870E-E GAMING WIFI Motherboard

They call this mainstream?

AMD's freshly launched 800-series chipsets are meant for motherboards that are to be paired with the new AMD Ryzen 9000 series processors. However, these new chipsets are part of the AM5 platform and AMD tends to keep backwards compatibility going for as long as they can. So, the new 800-series motherboards will support the Ryzen 7000, 8000 and 9000 processors. We're looking at the ASUS ROG STRIX X870E-E GAMING WIFI motherboard which is built around the top-tier X870E (E for Extreme) chipset and the ROG STRIX is their mainstream segment that sits above the value-driven TUF segment and alongside/above the PRIME segment. Since this board ships with an X870E chipset, it's one of the best motherboards you can get under the ROG STRIX segment. One look at the board and you'll notice that it's all decked out to the hilt. And ASUS being ASUS, always finds some way to slip in a little extra.

SPECIFICATIONS

We're going to be brief with the specifications here. The entire list can be found in our online review. Like we mentioned previously, the ASUS ROG STRIX X870E-E GAMING WIFI supports all Ryzen processors released on the AM5 socket including the 7000, 8000 and 9000 series. With Ryzen 8000 processors, the motherboard can RAM going up to 8400 MT/s while the 7000 and 9000 processors support up



to 8000 MT/s out of the box. You can obviously overclock to your heart's content. And since we're on the topic of RAM, it should be noted that 6400 MT/s is the sweet spot for Ryzen 9000 processors. That's when the MCLK and FCLK will run with a ratio of 1:1. Getting DDR5 memory kits with 6400 MT/s is a little difficult at the moment but with the launch of the new AMD 800 series chipsets, this should change over time. The mounting mechanism for the RAM modules is different with ASUS bringing in NitroPath which changes the way the pins are formed within the DRAM slots. The pins are now 39 per cent shorter and the latch

force has improved between 20 to 57 per cent.

Moving on to expansion ports, we have 1 PCIe 5.0 x16 slot attached to the CPU and another PCIe 5.0 x16 physical (but x4 electrical) from the chipset. There are five M.2 slots comprising three PCIe 5.0 slots from the processor and two PCIe 4.0 slots from the chipset. The M.2_2 and M.2_3 slots share bandwidth with the PCIe x16 G5 slot so using these M.2 slots will lead to the G5 slot running at x8 bandwidth.

One of the key changes from the AMD 600 to 800 series chipsets is the introduction of USB4 and the X870E has two USB4 Type-C ports running at 40 Gbps. Aside from that, we have one 20 Gbps Type-C port and ten 10 Gbps ports. The board also ships with Wi-Fi 7 and 5 GbE ethernet port. We'll elaborate more on the features as we look at each part of the motherboard.

VRM DESIGN

The ASUS ROG STRIX X870E-E GAMING WIFI uses a pretty standard VRM design for a board of this segment with 18 power phases for the Vcore and 2 phases for the Vsoc. In all, each phase can handle about 110 Amperes which is quite good for high-power consuming processors. If you're pairing this motherboard with the new Ryzen 9 9950X, then you'll need all the stability and phases you can get since it does consume upwards of 300 Watts under extreme multi-threaded load. Coming to the topology in place, the ASUS ROG STRIX X870E-E GAMING WIFI has a Teamed VRM layout which is better than Doublers owing to the fact that each phase gets a direct signal from the VRM controller and switching happens at the native frequency rather than the halved frequency in the case of doublers. The teamed solution leads to slightly higher ripple but there's lower transient voltage drop when switching from one phase to another, and therefore, there's a more stable power supply.